TEMPERATURE MONITOR DEV BOARD

[1] POWER

Power Input

File: power.kicad_sch

[2] MICROCONTROLLER

Microcontroller

File: mcu.kicad_sch

[3] DISPLAY

Display Interface

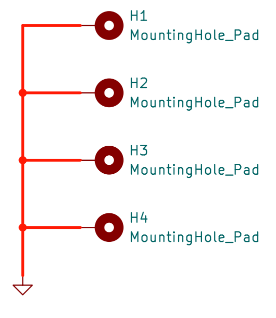
File: display.kicad_sch

[4] MISCELLANEOUS

Miscellaneous

File: miscellaneous.kicad_sch

[5] MOUNTING HOLES (M3)



M3 drill holes

[6] Fiducials

github.com/tsokomalusi

Sheet: /

File: Temperature_Monitoring_Dev_Board.kicad_sch

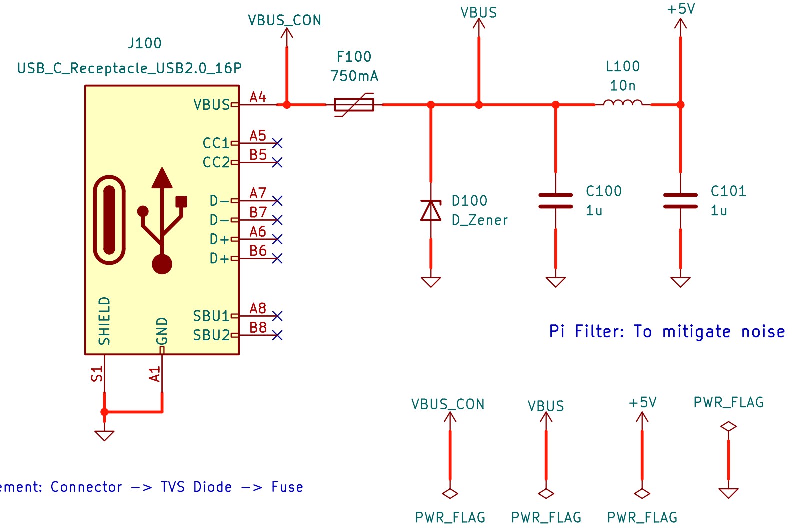
Title: Temperature Monitor Dev Board

Size: A3	Date: 2026-06-07	**Rev: 1.0**
KiCad E.D.A. 9.0.0		Id: 1/5

USB Type-C Connector

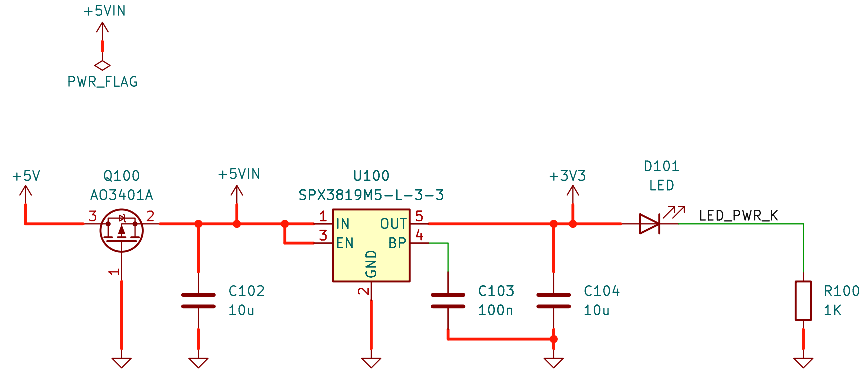
Input Protection: Fusing and ESD Protection

USB Type-C: Used for power-only not data.



Placement: Connector -> TVS Diode -> Fuse

Reverse Polarity Protection & LDO (+5V TO +3V3)



Note: Reverse polarity protection unnecessary for USB-Powered HW systems.
Only included here to satisfy PCB Design Challendge Requirements (PCB Skool Community)

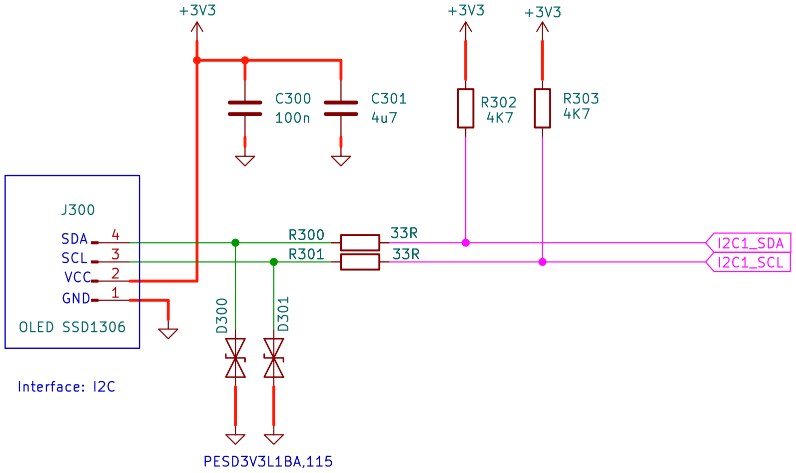
github.com/tsokomalusi

Sheet: /Power Input/
File: power.kicad_sch

Title: Temperature Monitor Dev Board

Size: A4	Date: 2026-06-07	**Rev: 1.0**
KiCad E.D.A. 9.0.0		Id: 2/5

OLED DISPLAY (SSD1306)



github.com/tsokomalusi

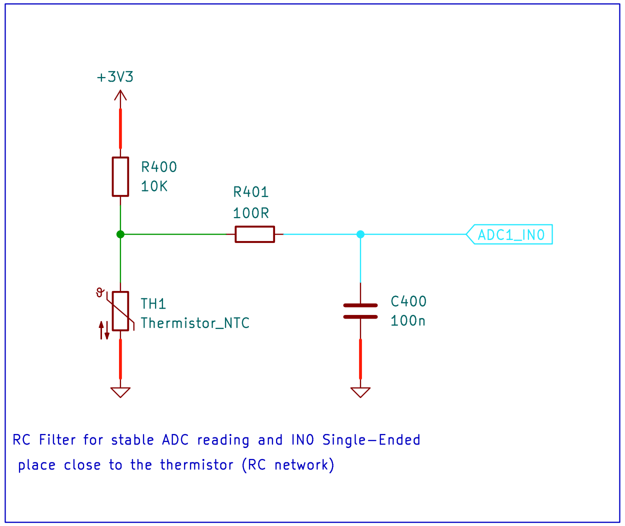
Sheet: /Display Interface/

File: display.kicad_sch

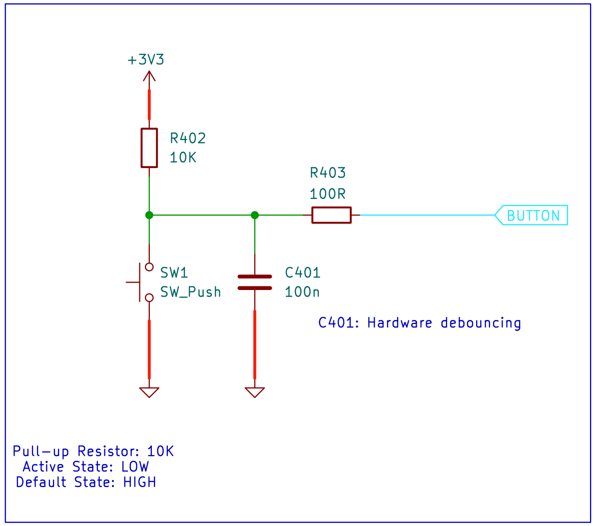
Title: Temperature Monitor Dev Board

Size: A4	Date: 2026-06-07	**Rev: 1.0**
KiCad E.D.A. 9.0.0		Id: 3/5

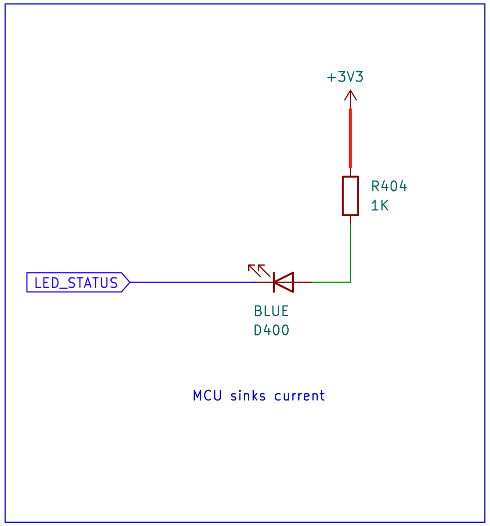
TEMPERATURE SENSING (NTC)



GPIO BUTTON



GPIO LED

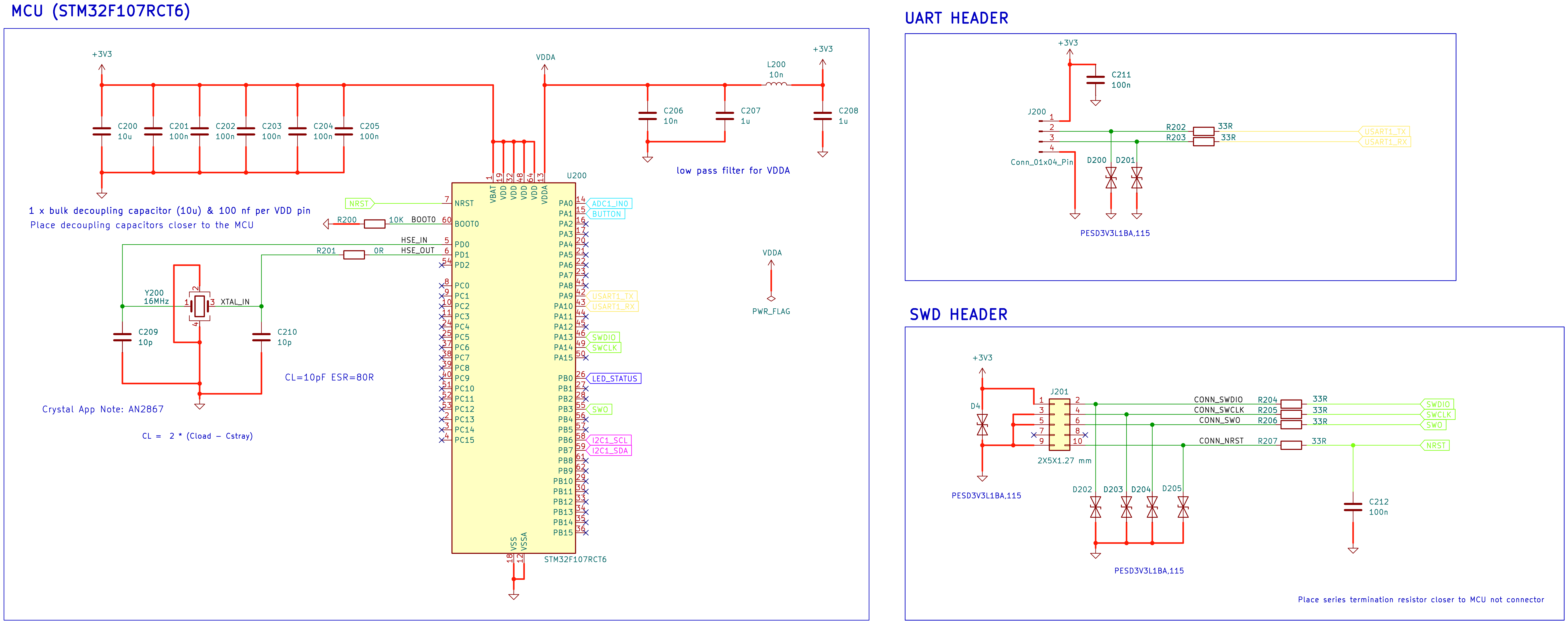


github.com/tsokomalusi

Sheet: /Miscellaneous/
File: miscellaneous.kicad_sch

Title: Temperature Monitor Dev Board

Size: A4	Date: 2026-06-07	**Rev: 1.0**
KiCad E.D.A. 9.0.0		Id: 5/5



github.com/tsokomalusi

Sheet: /Microcontroller/
File: mcu.kicad_sch

Title: Temperature Monitor Dev Board

Size: A3	Date: 2026-06-07	**Rev: 1.0**
KiCad E.D.A. 9.0.0		Id: 5/5